

SIMATS ENGINEERING

ASSESSMENT TOOL 2

COURSE CODE: CSA0908

COURSE NAME: Programming in Java

COURSE OUTCOME COVERED

CO2: Implement Collection Framework interfaces such as List, ArrayList, Set, Map, HashTable, and HashMap using iterators and generics to store, retrieve, and process groups of objects in web applications. (BL3, BL4)

Assessment Tool Weightage: 20%

QUESTIONS: 10

Total Marks: 50

ANNEXURE A

ASSIGNMENT 1

Code of Conduct:

I _ B.Deepthi-(192424269) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:B.Deepthi

1. Write a program to find the number of composite numbers in an array of elements

Sample Input:

Array of elements = {16, 18, 27, 16, 23, 21, 19}

Sample Output:

Number of Composite Numbers = 5

Test cases:

1. Array of elements = {26, 28, 37, 26, 33, 31, 29}
2. Array of elements = {1.6, 1.8, 2.7, 1.6, 2.3, 2.1, .19}
3. Array of elements = {0, 160, 180, 270, 160, 230, 210, 190, 0}
4. Array of elements = {200, 180, 180, 270, 270, 270, 190, 200}
5. Array of elements = {100, 100, 100, 100, 100, 100, 100, 100}

OUTPUT:

The screenshot displays the SIMATS Online Programming Lab interface. The header includes the SIMATS Engineering logo, the lab name, and a user profile for BOLLEPALLI DEEPTHI. The main content area shows a problem titled "Number of Composite Numbers = 5" with five test cases. The "Your Code" section contains a Java program that uses a Scanner to read an array of integers and counts the composite numbers. The "Input" section shows the test case values: 7, 26, 28, 37. The "Output" section displays the result: "Number of Composite Numbers = 4". The "Visible Test Cases" section shows "Test Case 1".

Number of Composite Numbers = 5

Test Cases:

- 1,{26,28,37,26,33,31,29}
- 2,{1.6,1.8,2.7,1.6,2.3,2.1,.19}
- 3,{0,160,180,270,160,230,210,190,0}
- 4,{200,180,180,270,270,270,190,200}
- 5,{100,100,100,100,100,100,100,100}

Your Code

```
1 import java.util.Scanner;
2 public class Composite {
3     public static void main(String[] args) {
4         Scanner sc = new Scanner(System.in);
5         int n = sc.nextInt();
6         int[] arr = new int[n];
7         for (int i = 0; i < n; i++) {
8             arr[i] = sc.nextInt();
9         }
10        int count = 0;
11        for (int i = 0; i < n; i++) {
12            if (isComposite(arr[i])) {
13                count++;
14            }
15        }
16        System.out.println("Number of Composite Numbers = " + count);
17    }
18    private boolean isComposite(int num) {
19        if (num < 2) return false;
20        for (int i = 2; i <= Math.sqrt(num); i++) {
21            if (num % i == 0) return true;
22        }
23        return false;
24    }
25 }
```

Input

7
26
28
37
...

Output

Number of Composite Numbers = 4

Visible Test Cases

Test Case 1

2. Write a program to print the numbers from M to N by skipping K numbers in between?

Sample Input:

M = 50

N = 100

K = 7

Sample Output:

50, 58, 66, 74,

Test cases:

1. M = 15, N = 05, K = 02
2. M = 25, N = 50, K = 04
3. M = 15, N = 100, K = -02
4. M = 0, N = 0, K = 2
5. M = 200, N = 200, K = 50

OUTPUT:

The screenshot shows the SIMATS Online Programming Lab interface. The header includes the SIMATS ENGINEERING logo, the title "SIMATS Online Programming Lab", and the user "BOLLEPALLI DEEPTHI". The main content area displays a problem statement: "Write a program to print the numbers from M to N by skipping K numbers in between. Sample: M=50 N=100 K=7". Below this, there is a "Your Code" section with a Java program that uses a Scanner to read M, N, and K, and a while loop to print numbers from M to N, skipping K numbers. The "Input" section shows the values 50, 100, and 7. The "Output" section shows the sequence 50 58 66 74 82 90 98. On the left, there is a sidebar with a list of questions (Q1 to Q5) and their marks. At the bottom, there are "Run" and "Save" buttons.

```
1 import java.util.Scanner;
2 public class Skip{
3     public static void main(String[] args) {
4         Scanner sc = new Scanner(System.in);
5         int m = sc.nextInt();
6         int n = sc.nextInt();
7         int k = sc.nextInt();
8         int i = m;
9         while (i <= n) {
10             System.out.print(i + " ");
11             i = i + k + 1;
12         }
13     }
14 }
```

3. Write a program to arrange the letters of the word alphabetically in reverse order

Sample Input:

Enter the word: MOSQUE

Sample Output:

Alphabetical Order: U S Q O M E

Test Case:

1. HYPOTHECATION
2. MATRICULATION
3. MANIPULATION

OUTPUT:

The screenshot shows the SIMATS Online Programming Lab interface. The browser address bar indicates the URL: `simatslab.saveetha.com/practice_app/classactivity/attempt?assignment_id=458`. The page header includes the SIMATS ENGINEERING logo, the title "SIMATS Online Programming Lab", and the user name "BOLLEPALLI DEEPTHI". The main content area displays a programming task: "Write a program to arrange the letters of the word alphabetically in reverse order." with a sample input "MOSQUE" and the expected output "U S Q O M E". The "Your Code" section contains a Java program that reads a word, converts it to uppercase, sorts the characters, and prints them in reverse order. The "Input" field contains "MOSQUE", and the "Output" field shows "U S Q O M E". The "Visible Test Cases" section lists three test cases, each with a radio button. The "Run" and "Save" buttons are at the bottom.

```
1 import java.util.Arrays;
2 import java.util.Scanner;
3 public class Reverse {
4     public static void main(String[] args) {
5         Scanner sc = new Scanner(System.in);
6         String word = sc.nextLine();
7         word = word.toUpperCase();
8         char[] letters = word.toCharArray();
9         Arrays.sort(letters);
10        StringBuilder result = new StringBuilder();
11        for (int i = letters.length - 1; i >= 0; i--) {
12            result.append(letters[i]);
13            if (i != 0) {
14                result.append(" ");
15            }
16        }
17        System.out.println(result.toString());
18    }
19 }
```

4. Write a program using a choice to check

Case 1: Given string is a palindrome or not

Case 2: Given number is palindrome or not

Sample Input:

Case = 1

String = MADAM

Sample Output:

Palindrome

Test cases:

1. MONEY
2. 5678765
3. MALAY12321ALAM
4. MALAYALAM
5. 1234.4321

OUTPUT :

The screenshot displays the SIMATS Online Programming Lab interface. The header includes the SIMATS Engineering logo, the lab name, and the user's name (BOLLEPALLI DEEPTHI). The main area is divided into three sections: 'Your Code', 'Input', and 'Output'. The 'Your Code' section contains a Java program that checks for string and number palindromes. The 'Input' section shows the user's input (2, 524). The 'Output' section shows the program's output (1. String Palindrome, 2. Number Palindrome, Enter number: Not Palindrome). The 'Visible Test Cases' section lists three test cases.

```
1 import java.util.*;
2 public class Main {
3     public static void main(String[] args) {
4         Scanner sc = new Scanner(System.in);
5         System.out.println("1. String Palindrome");
6         System.out.println("2. Number Palindrome");
7         int ch = sc.nextInt();
8         if (ch == 1) {
9             sc.nextLine();
10            System.out.print("Enter string: ");
11            String s = sc.nextLine();
12            String r = new StringBuilder(s).reverse().toString();
13            System.out.println(s.equalsIgnoreCase(r) ? "Palindrome" : "Not Palindrome");
14        }
15        else if (ch == 2) {
16            System.out.print("Enter number: ");
17            int n = sc.nextInt();
18            int temp = n, rev = 0;
19            while (temp != 0) {
20                rev = rev * 10 + (temp % 10);
21                temp = temp / 10;
22            }
23            System.out.println(n == rev ? "Palindrome" : "Not Palindrome");
24        }
25        else {
26            System.out.println("Invalid choice");
27        }
28    }
29 }
```

Input:

2
524

Output:

1. String Palindrome
2. Number Palindrome
Enter number: Not Palindrome

Visible Test Cases:

Test Case 1
Test Case 2
Test Case 3

5. Java Program to Find the First Non-Repeated Character in a String.

Case 1: Find the first non-repeated character in a given string.

Case = 1

String = SWISS

First Non-Repeated Character = W

Test Cases

1. HELLO
2. SWISS
3. PROGRAMMING
4. AABBCDD
5. SUCCESS

OUTPUT:

The screenshot displays the SIMATS Online Programming Lab interface. The browser address bar shows the URL: `simatslab.saveetha.com/practice_app/classactivity/attempt?assignment_id=458`. The page header includes the SIMATS ENGINEERING logo, the text "SIMATS Online Programming Lab", and the user's name "BOLLEPALLI DEEPTHI" with a user ID "192424269RA".

On the left sidebar, there is a list of questions:

- Q2: Skip K Numbers (5 marks)
- Q3: Reverse Alphabetical Order (5 marks)
- Q4: Palindrome Check (5 marks)
- Q5: First Non-Repeated Character (5 marks)

The main area shows the "Your Code" section with the following Java code:

```
1 import java.util.*;
2 public class NonRepeatedCharacter {
3     public static void main(String[] args) {
4         Scanner sc = new Scanner(System.in);
5         String s = sc.nextLine();
6         for (int i = 0; i < s.length(); i++) {
7             int cnt = 0;
8             for (int j = 0; j < s.length(); j++) {
9                 if (s.charAt(i) == s.charAt(j)) cnt++;
10            }
11            if (cnt == 1) {
12                System.out.print(s.charAt(i));
13                return;
14            }
15        }
16        System.out.print(-1);
17    }
18 }
```

The "Input" section shows the text "madam". The "Output" section shows the character "d". The "Visible Test Cases" section lists three test cases, each with a radio button:

- Test Case 1
- Test Case 2
- Test Case 3

At the bottom, there are "Run" and "Save" buttons.